



**Faculty of Engineering & Technology Electrical & Computer Engineering
Department
Circuits and Electronics Laboratory - ENEE2103
Exp5 : Filters
Report#2**

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Abstract:

This experiment aims to introduce some basic concepts of filtering using different filtering circuits like passive filters (first order and second order) and active filters (circuits that incorporate an OPAMP). As well as the behavior of each one's output, by evaluating the amplitude and frequency of the outputs with an oscilloscope and a digital multimeter.

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1.Theory

Electric filters have many applications and are extensively used in many signal processing circuits. It is used for choosing or eliminating signals of selected frequency in a complete spectrum of a given input. So the filter is used for allowing signals of chosen frequency pass through it or eliminate signals of chosen frequency passing through it . there are many types of filters available and they are differentiated in many ways

1.1 Passive Filters

First order filters :

A low-pass filter is a filter that passes signals with a frequency lower than a selected cutoff frequency and attenuates signals with frequencies higher than the cutoff frequency .

Passive filters are made up of passive components such as resistors, capacitors and inductors and have no amplifying elements (transistors, op-amps, etc) so have no signal gain, therefore their output level is always less than the input , Simple First-order passive filters (1st order) can be made by connecting together a single resistor and a single capacitor in series across an input signal, (V_{IN}) with the output of the filter, (V_{OUT}) taken from the junction of these two components.

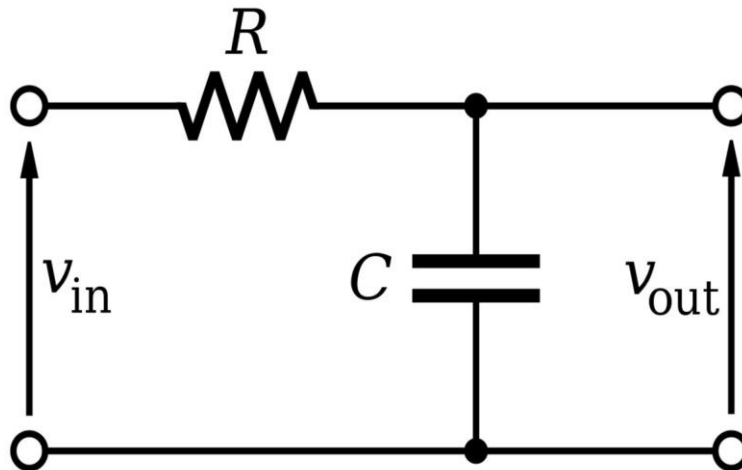


Figure 1.1.1 : First Order Circuit connection

Second order filters :

Band pass filters allow only a required band of frequencies to pass, while rejecting signals at all frequencies above and below this band. This particular design is called a T filter because of the way the components are drawn in a schematic diagram. The T filter consists of three elements, two series-connected LC circuits between input and output, which form a low impedance path to signals of the required frequency, but have a high impedance to all other frequencies.

Additionally, a parallel LC circuit is connected between the signal path (at the junction of the two series circuits) and ground to form a high impedance at the required frequency, and a low impedance at all others. Because this basic design forms only one stage of filtering it is also called a 'first order' filter. Although it can have a reasonably narrow pass band, if sharper cut off is required, a second filter may be added at the output of the first filter, to form a 'second order' filter.

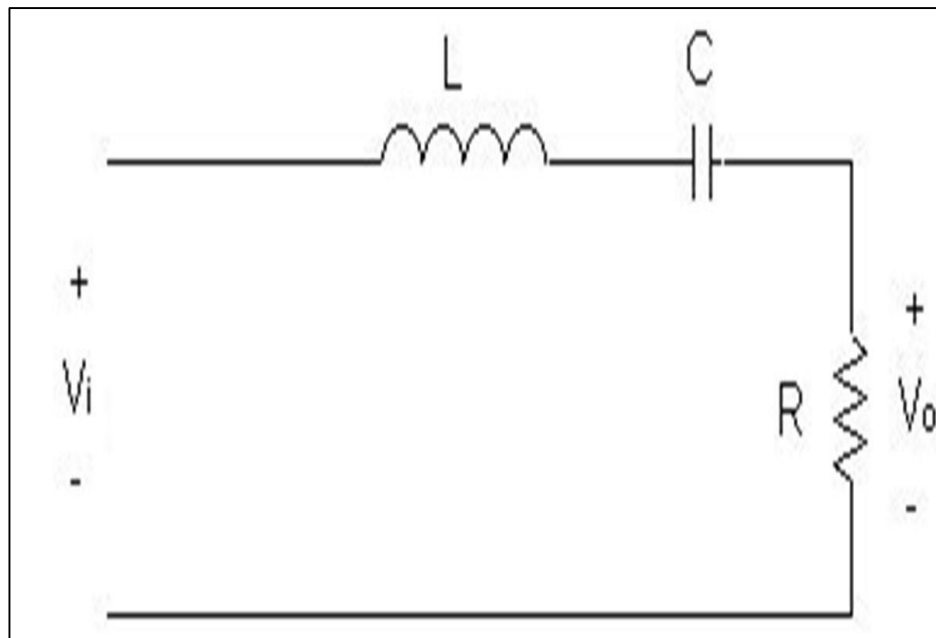


Figure 1.1.2 : Band pass Filter Circuit Connection

1.2 Active Filters

Its principle of operation and frequency response is exactly the same as those for the previously seen passive filter, the only difference this time is that it uses an op-amp for amplification and gain control. The simplest form of a low pass active filter is to connect an inverting or non-inverting amplifier .

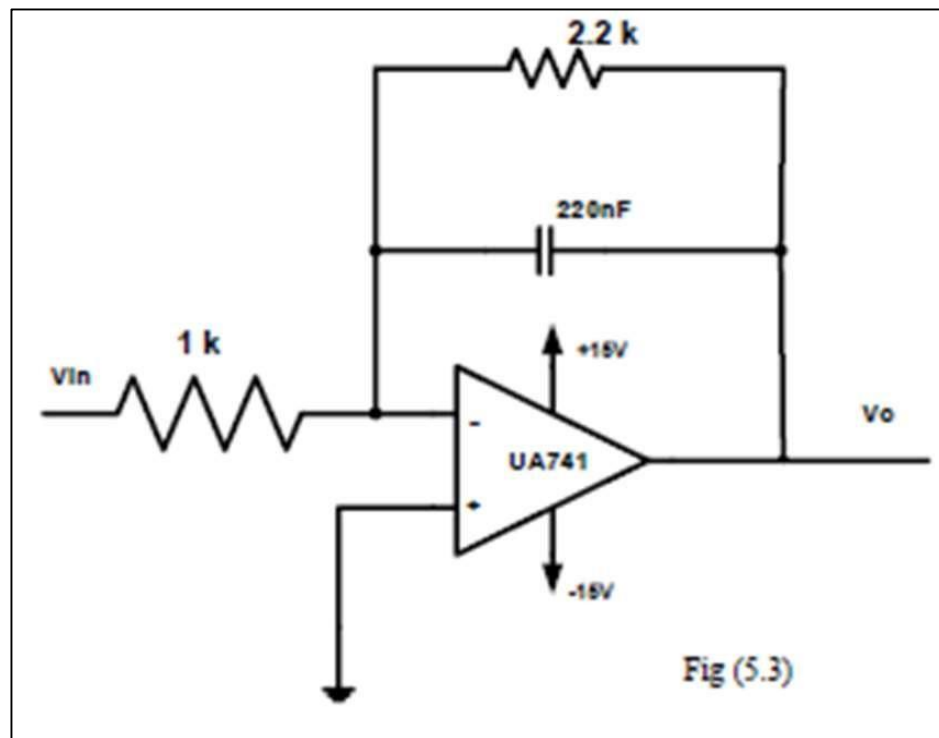


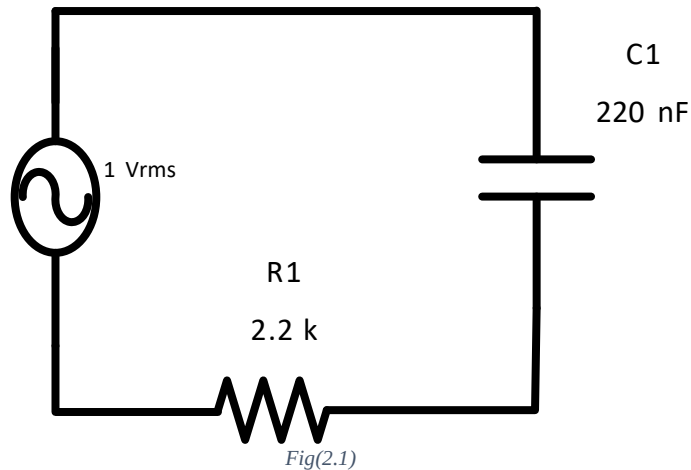
Figure 1.2.1 : Band Pass Filter

2.Procedure

2.1 Passive filters:

2.1.1 First order circuits:

1. The circuit of Fig (2.1) is connected .



2. The signal generator is set to generate a sinusoidal waveform with 1 Vrms and $f = 20 \text{ Hz}$ and the output voltage on the capacitor is measured using the DMM.
3. The phase shift between the **circuit current and the input** voltage are determined using the oscilloscope.
4. V_i , V_R and V_C are measured using DMM .
5. The value of cut-off frequency (f_c) is measured by finding the max value of Voltage across the capacitor V_{C_max} (use DMM) first, then f_c is the frequency for which $V_C = 0.707 V_{C_max}$. (note: value of $f_c = 1/(2R_{eq}C)$).

2.1.2 Second Order Filters:

1. The circuit of Fig (2.2) is connected .

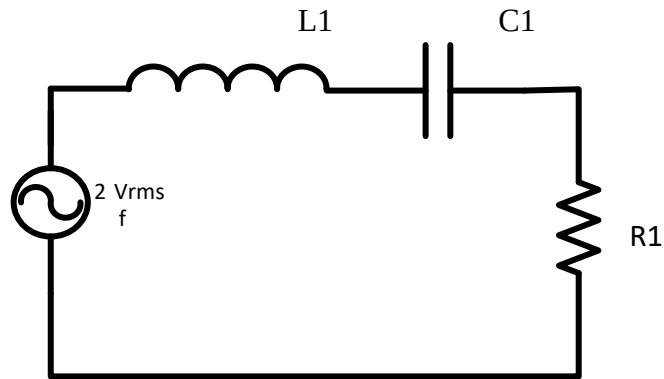


Fig (2.2)

2. The component values **R1=1 k**, **L1=100 mH** and **C1=470nF** are set .
3. The signal generator is set with 2 volts (RMS) and a frequency value that equals the resonance frequency (f_0) measured experimentally and verified theoretically.
4. The voltages across each component: V_R , V_L , V_C measure using the DMM.

2.2 Active filters:

1. The circuit of Fig (2.3) is connected .

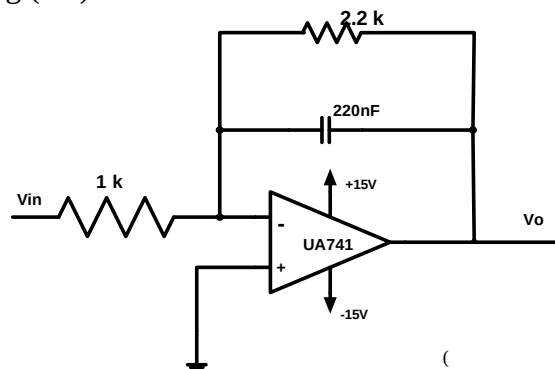


Fig (2.3)

2. The expected value of cut-off frequency f_c is calculated.

3. The signal generator is set with 3 volts amplitude and $f = f_c$ and the input and output voltage are measured using the DMM.

3. Data & Results :

3.1 Passive Filters

3.1.1 First Order Filter

In this part the circuit was connected as circuit shown in figure 3.1 then using DMM , V_c , V_r were measured and there values are shown in the data sheet . Using the oscilloscope , the phase shift between the current and the input voltage was determined in different frequencies .

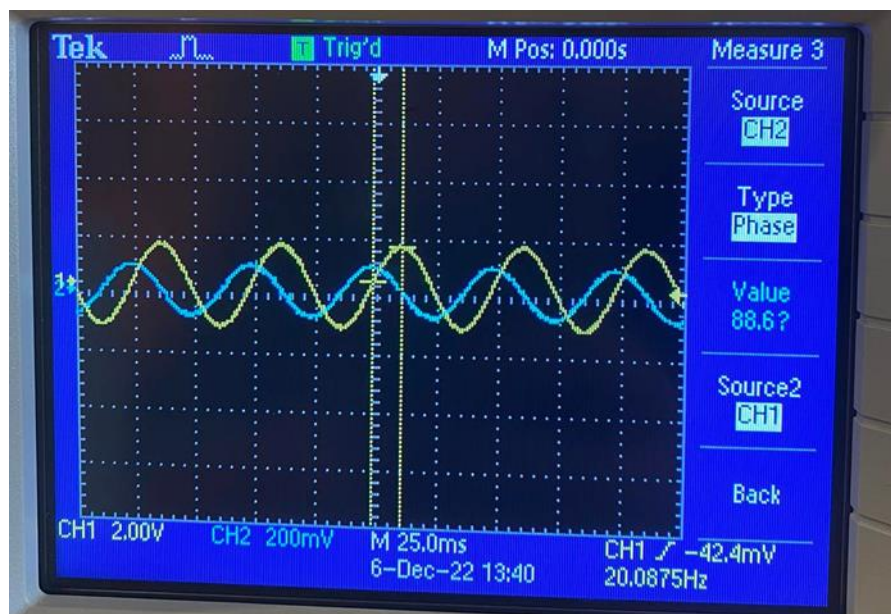


Figure 3.1 first order output

Discussion :

Capacitor in this circuit worked as low pass filter, hence we can see it lets the low frequencies pass and rejects the high frequencies. Also we notice that the phase shift between V_{in} and V_c in lower frequencies was very low but in the high frequencies the phase shift was higher . V_{in} was leading in both cases. On the other hand, the resistance worked as High pass filter , first V_r was nearly zero but as we increase the frequency , V_r also was getting bigger until it almost reached V_p . for the phase shift between them was high in the low frequencies but in the high frequencies it went to almost zero .

Table 3.1.1 Data for low Pass filter

SET		MEASURE			CALCULATE		
Frequency	V _{in} (V _{rms})	V _R (V _{rms})	V _C (V _{rms})	θV-θI	log f	20 log (V _R /V _{in})	20 log (V _C /V _{in})
20Hz	1V _{rms}	0.065	1.043	-88.6	1.30	-23.7	0.365
0.1fc	1V _{rms}	0.098	1.035	-85.5	1.49	-20.17	0.298
0.5fc	1V _{rms}	0.442	0.934	-64.4	2.18	-7.09	-0.593
F _c	1V _{rms}	0.705	0.742	-44.2	2.49	-3.03	-2.591
2fc	1V _{rms}	0.903	0.476	-25.1	2.79	-0.886	-6.447
4fc	1V _{rms}	0.982	0.259	-14.7	3.09	-0.15	-11.73
5fc	1V _{rms}	0.993	0.209	-13.4	3.18	-0.06	-13.59
10fc	1V _{rms}	1.005	0.106	-8.31	3.49	0.04	-19.49
12fc	1V _{rms}	1.004	0.088	-7.88	3.56	0.034	-21.1
14fc	1V _{rms}	1.003	0.076	-6.71	3.63	0.026	-22.38
16fc	1V _{rms}	1.003	0.068	-3.89	3.69	0.026	-23.34
20fc	1V _{rms}	1.007	0.056	-1.84	3.79	0.06	-25.03

- Calculations :

Value of f_c is obtained by solving the following quadratic equation :

$$LC*W^2 - RC*W - 1 = 0$$

➤ $F_c = 328.9\text{Hz}$

- Using Excel :

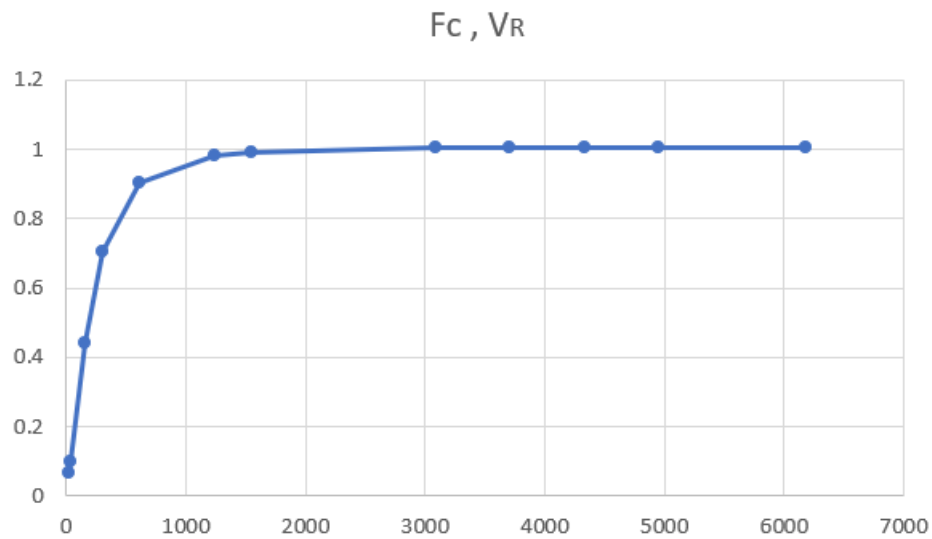


Figure 3.1 : F_c, V_R

- High pass filter
- -3db cut-off frequency $F=309.3\text{Hz}$

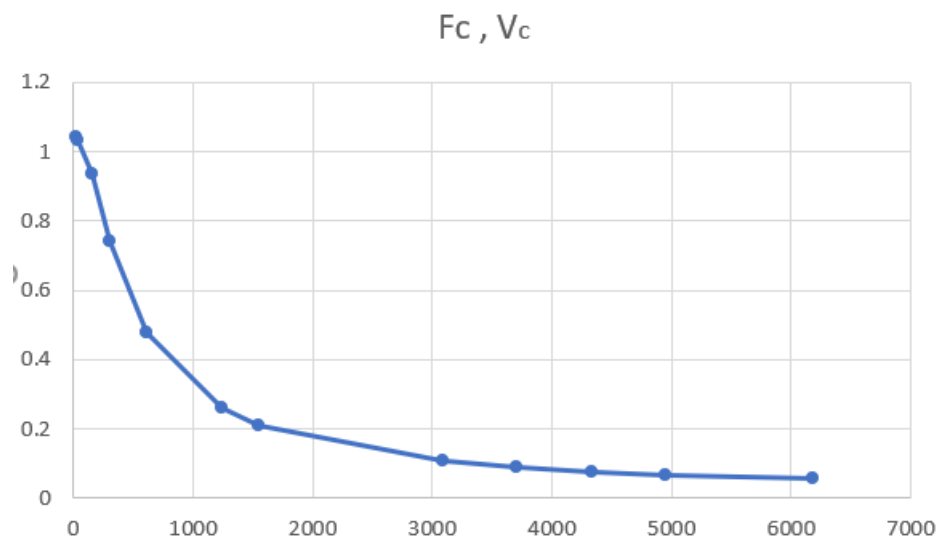


Figure 3.1 : F_c, V_c

- Low pass filter

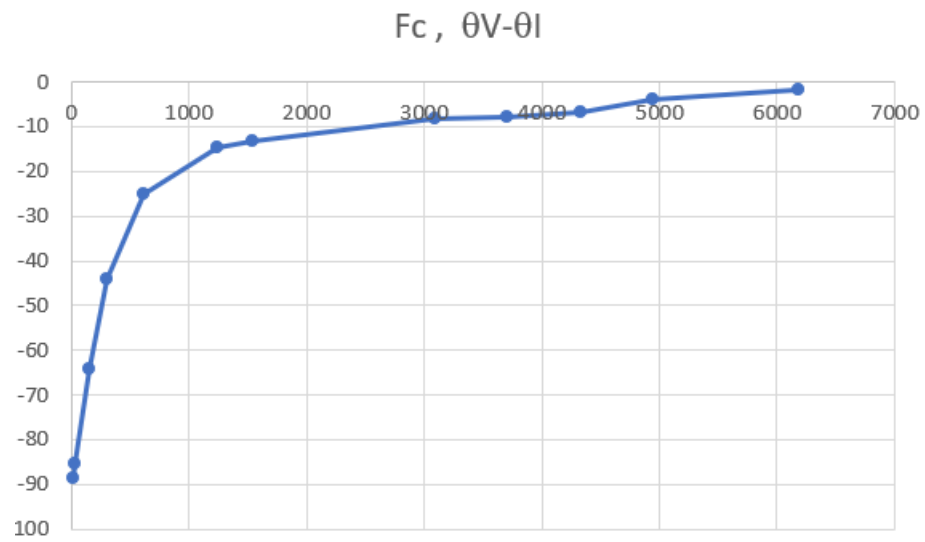


Figure 3.1: F_c , phase shift

The results were validated using PSpice : (at all section)

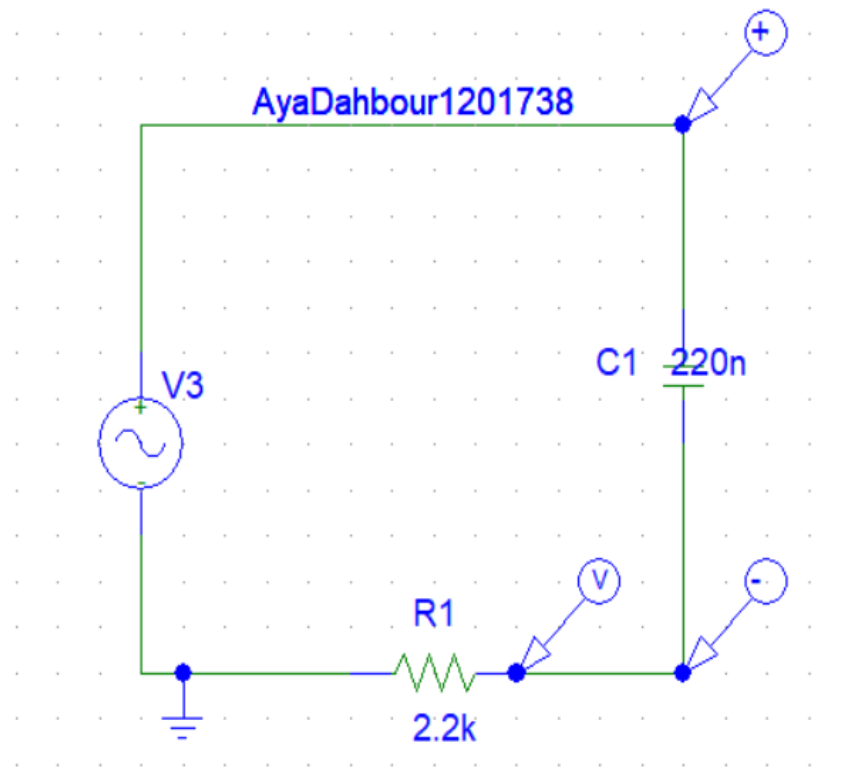


Figure 1 Figure 3.1:circuit 1 pspice

The green output of VR1 and the red of VC1 :

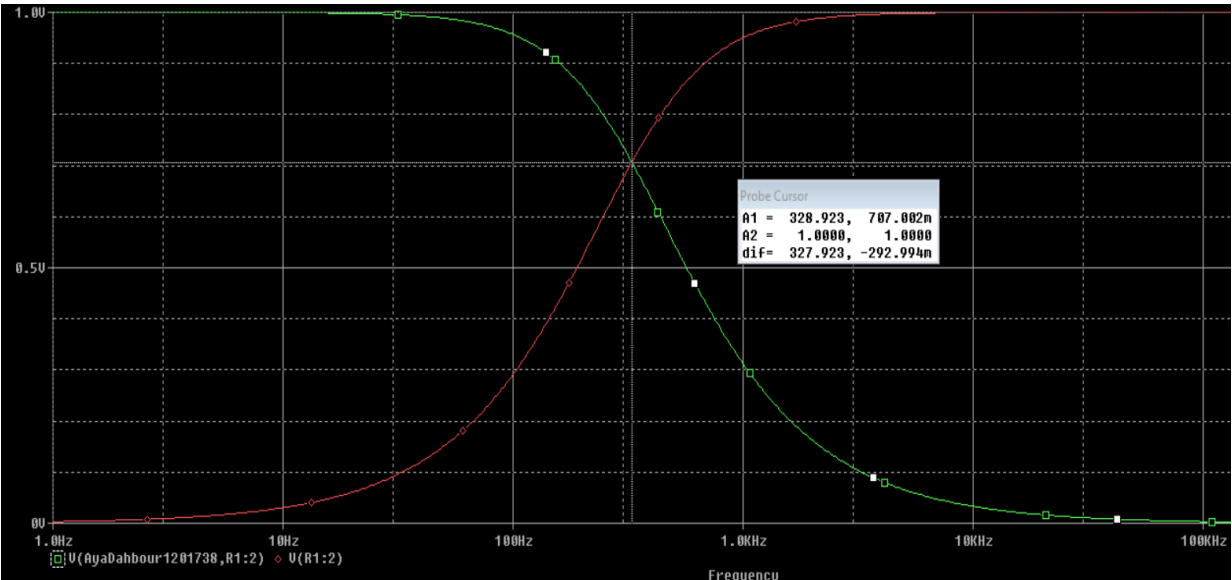
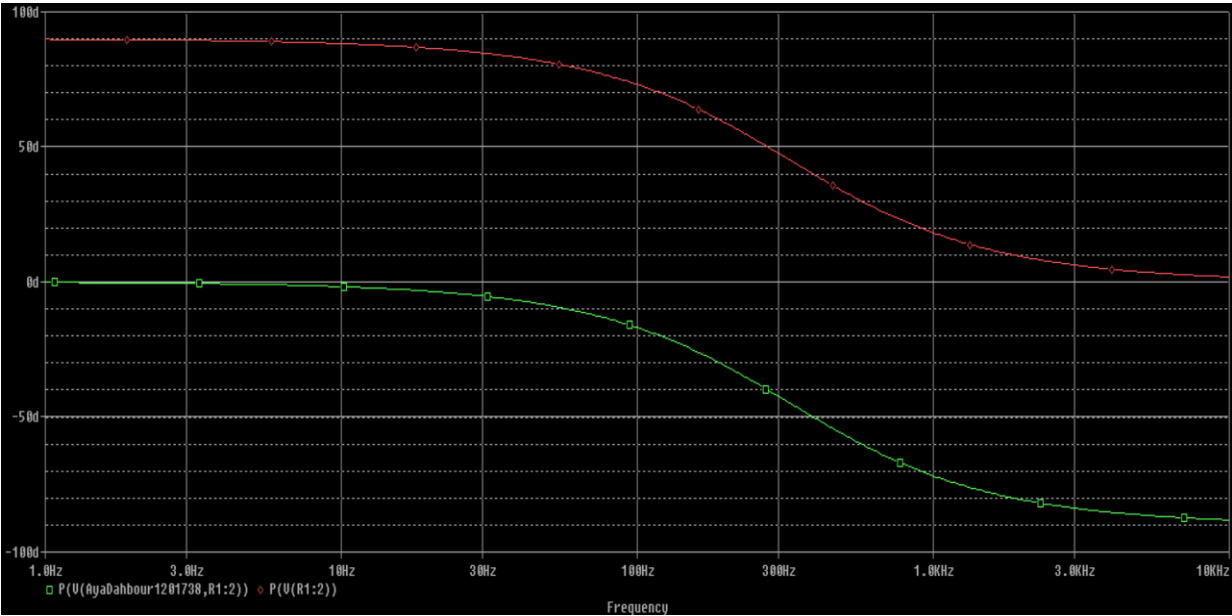


Figure 2 Figure 3.1: output VR1 & VC1 pspice

phase of VR and Vc :



3.1.2 Second Order Filter

First circuit was connected as shown in figure 3.1.2 then using DMM V_c , V_r , V_l were measured and there values are shown in the data sheet.

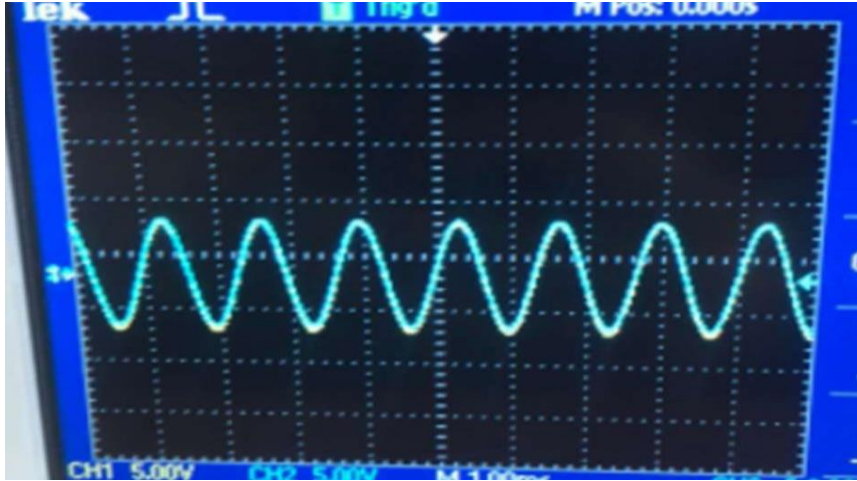


Figure 3.1.2: second order output

Discussion :

The resistance worked as filter for the frequencies between f_{c1} and f_{c2} passed through it with no big losses , on the other hand the inductor and the capacitor together work as band stop filter that the frequencies between f_{c1} and f_{c2} stopped went through with no big losses .

Table 3.1.2 : Data for Band Pass filter

SET		MEASURE		CALCULATE		
Frequency	V _{in} (V _{rms})	V _R (V _{rms})	(V _C +V _L)(V _{rms})	log f	20 log (V _R /V _{in})	20 log (V _C +V _L /V _{in})
0.1 fc1	2V _{rms}	0.026	2.032	1.439	-37.72	0.137
0.3fc1	2V _{rms}	0.492	1.998	1.916	-12.18	-8.6*10 ⁻³
0.6fc1	2V _{rms}	0.933	1.809	2.217	-6.62	-0.871
fc1	2V _{rms}	1.375	1.456	2.439	-3.254	-2.757
290	2V _{rms}	1.422	1.399	2.462	-2.962	-3.1
310	2V _{rms}	1.478	1.331	2.491	-2.627	-3.53
350	2V _{rms}	1.583	1.186	2.544	-2.03	-4.538
0.5fo	2V _{rms}	1.613	1.138	2.56	-1.867	-4.897
Fo	2V _{rms}	1.955	0.032	2.861	-0.197	-35.91
2fo	2V _{rms}	1.654	1.115	3.162	-1.649	-5.075
1700	2V _{rms}	1.51	1.326	3.23	-2.441	-3.569
fc2	2V _{rms}	1.386	1.474	3.29	-3.185	-2.65
2fc2	2V _{rms}	0.78	1.902	3.595	-8.17	-0.436
5fc2	2V _{rms}	0.279	2.059	3.993	-17.1	0.252
10fc2	2V _{rms}	0.041	2.055	4.294	-33.76	0.235
12fc2	2V _{rms}	0.005	2.044	4.373	-52.04	0.189
14fc2	2V _{rms}	0.025	2.031	4.44	-38.06	0.133
17fc2	2V _{rms}	0.071	2.009	4.524	-28.99	0.038
20fc2	2V _{rms}	0.109	1.985	4.595	-25.27	-0.065

Calculations :

$$f_{Resonance} = \frac{1}{2\pi\sqrt{LC}} \quad f_{Resonance} = 734.5 \text{ , experimentally } f_o = 726.6\text{Hz}$$

$$f_{c1} = \frac{1}{2\pi} \times \left(-\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} \right)$$

$$\Rightarrow F_{c1} = 288\text{Hz , experimentally } F_{c1} = 275\text{Hz}$$

$$\Rightarrow F_{c2} = \frac{1}{2\pi} \times \left(\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} \right)$$

$$\Rightarrow F_{c2} = 1879.4\text{Hz experimentally } F_{c2} = 1969\text{Hz}$$

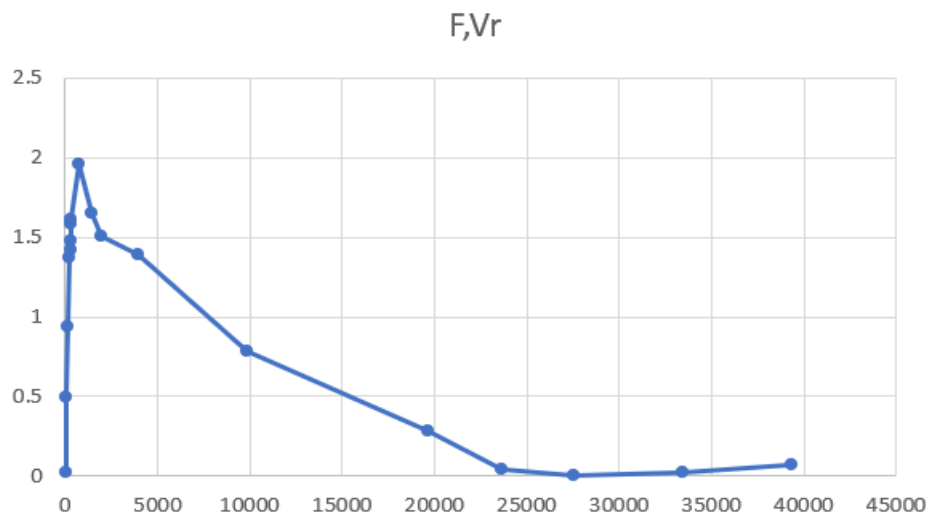


Figure 3.1: F, V_r

- Band pass filter

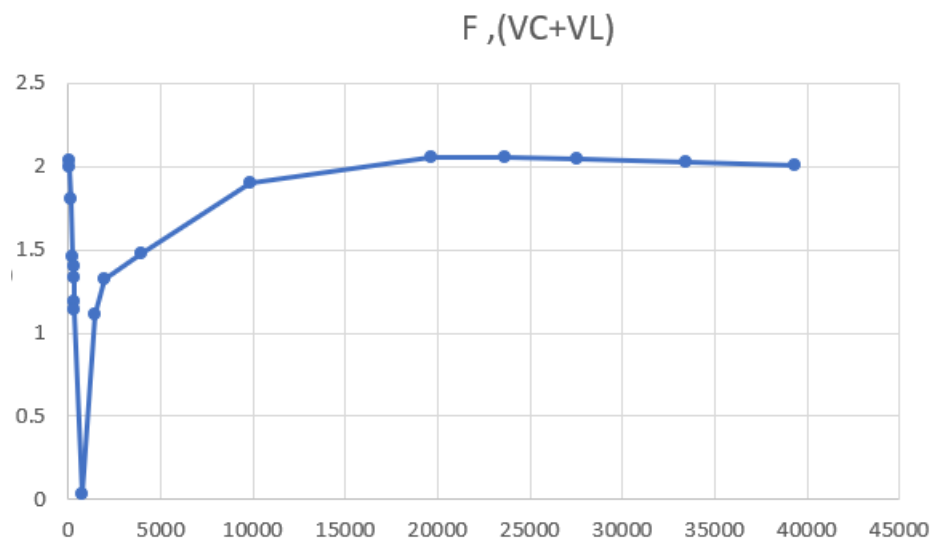


Figure 3.1 : $F, (V_c + V_l)$

- Band stop filter

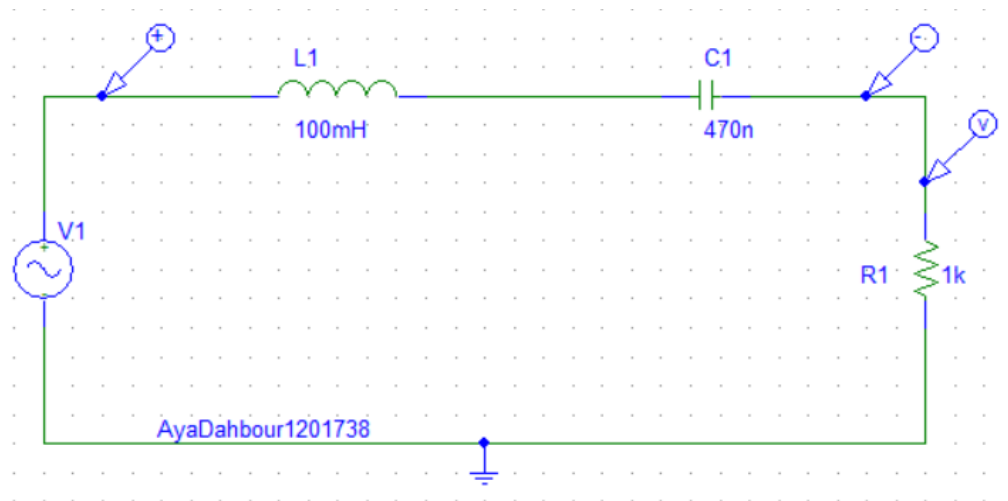


Figure 4 Figure 3.1 : circuit 2 pspice

- Amplitude of VR and (Vc+VL) pspice:

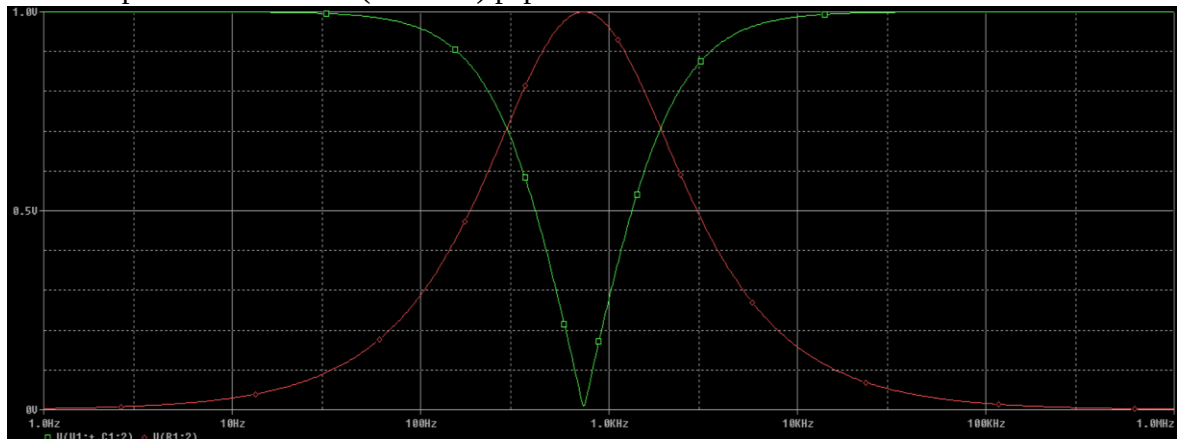


Figure 5 Figure 3.1 : amplitude of VR and (Vc+VL) pspice

- Phase of VR and (Vc+VL) pspice:

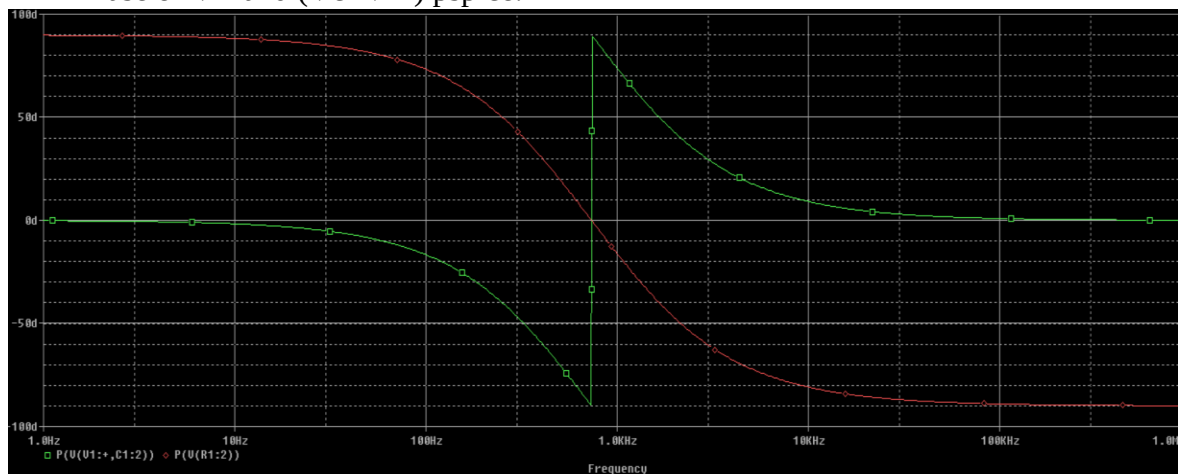


Figure 6 Figure 3.1 : phase of VR and (Vc+VL) pspice

3.2 Active Filters

First circuit was connected as shown in figure 3.2 then using DMM V_o was measured and there values are shown in the data sheet .



Figure3.2.2 output for Active Low pass filter

$$F_c = 336.5\text{Hz}$$

Discussion :

We noticed from table 3.2 that circuit is a low pass filter which lets the low frequencies pass and rejects the high frequencies.. If we compared part a1 with this part, we can notice that output voltage in this part acts like voltage of resistor in the first part as both are low pass filter. But one of them are passive filter (uses passive components) and the other one is active and the voltage is amplified.

Table 3.2 : Data for low Pass filter

SET	MEASURE		CALCULATE	
Frequency	V _{in} (V _{rms})	V _o (V _{rms})	log f	20 log (V _o /V _{in})
0.1 fc	1.04	2.26	1.526	6.741
0.2fc	1.04	2.19	1.828	6.468
0.4fc	1.04	2.066	2.129	5.961
0.8fc	1.04	1.719	2.43	4.364
0.9fc	1.04	1.632	2.48	3.913
Fc	1.04	1.594	2.526	3.709
1.3fc	1.04	1.327	2.64	2.116
1.5fc	1.04	1.205	2.703	1.279
2fc	1.04	0.961	2.828	-0.686
4fc	1.04	0.533	3.129	-5.806
6fc	1.04	0.328	3.305	-10.02
10fc	1.04	0.217	3.526	-13.61
15fc	1.04	0.144	3.703	-17.17
20fc	1.04	0.112	3.828	-19.35

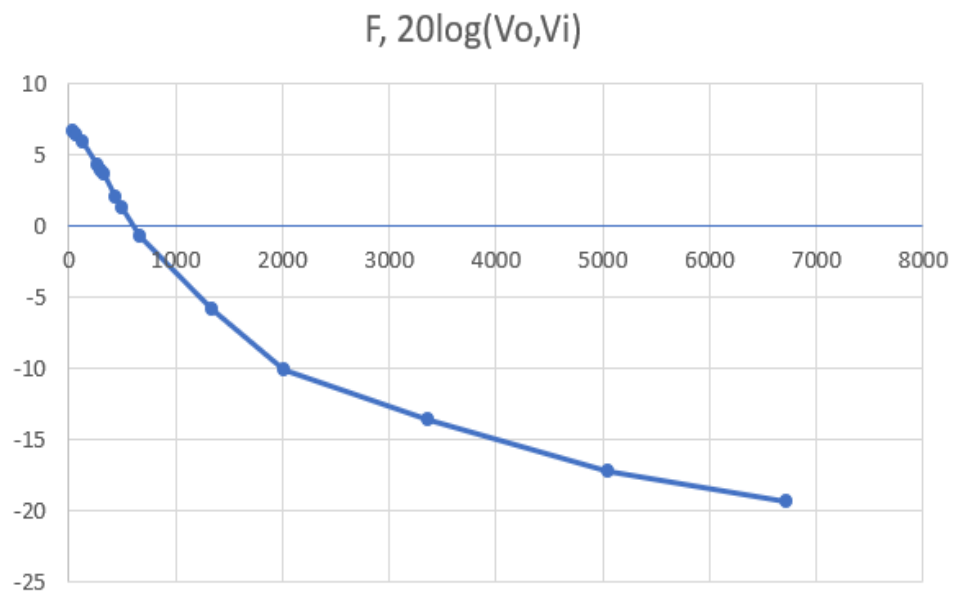


Figure 3.2 : F, V_o

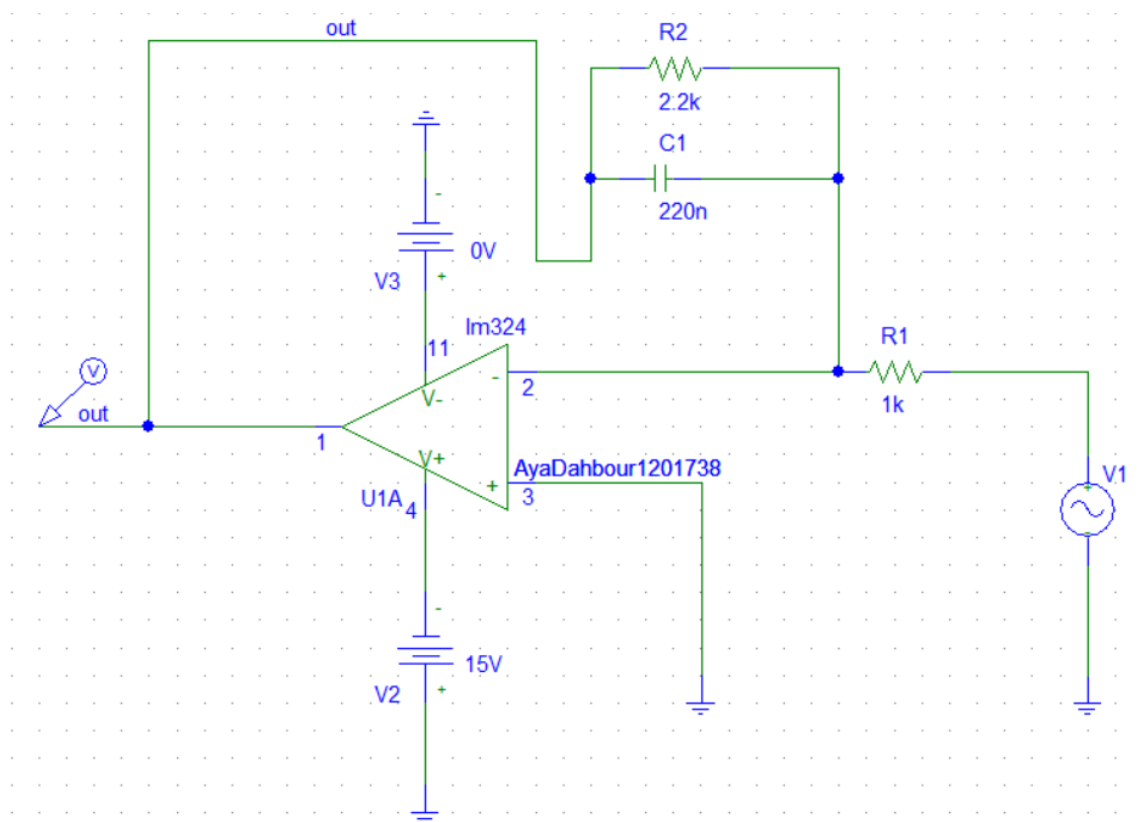


Figure 7Figure 3.2 : circuit 3 pspice

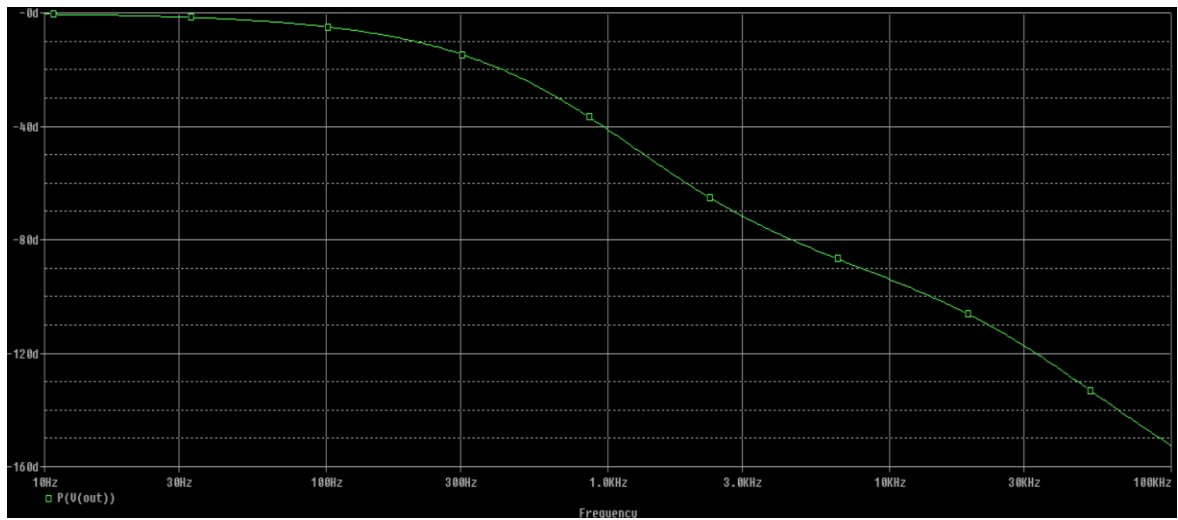


Figure 8Figure 3.2: Vo pspice

4. Conclusion:

in this experiment, we learned about filters, their different types like active and passive filters, also we saw the effect of changing multiple parameters on these filters circuits, especially the frequency and measured the voltages magnitudes in different frequencies to determine the type of filter (Low, High, pass) and found the cut off frequencies for different filters .

5. References

[1] lab manual .

[2] https://www.electronics-tutorials.ws/filter/filter_5.html

accessed on 7/12/2022 at 10:30 pm

[3] <https://www.dummies.com/education/science/science-electronics/create-band-pass-and-band-reject-filters-with-rlc-series-circuits>

accessed on 8/12/2022 at 10:30 am

[4] https://learnabout-electronics.org/ac_theory/filters81.php

accessed on 8/12/2022 at 11:06 am

[5] https://www.electronics-tutorials.ws/filter/filter_5.html

accessed on 8/12/2022 at 11:32 am

Table 5.1

SET		MEASURE			CALCULATE		
Frequency	V _{in} (Vrms)	V _R (Vrms)	V _C (Vrms)	θ _{V-θ_I}	log f	20 log (V _R /V _{in})	20 log (V _C /V _{in})
20 Hz	1Vrms	0.065	1.000	-87.5			
0.1 fc	1Vrms	0.098	1.000	-85.5			
0.5 fc	1Vrms	0.442	0.030	-21.4			
F _c	1Vrms	0.705	0.002	-44.2			
2F _c	1Vrms	0.903	0.000	-55.1			
4F _c	1Vrms	0.982	0.000	-10.2			
5F _c	1Vrms	0.993	0.000	-13.4			
10F _c	1Vrms	1.005	0.000	-18.5			
12F _c	1Vrms	1.004	0.000	-23.8			
14F _c	1Vrms	1.003	0.000	-27.1			
16F _c	1Vrms	1.003	0.000	-28.9			
20F _c	1Vrms	1.007	0.000	-31.8			

$$R = 2.173$$

$$C = 225 \text{ S}$$

$$f_c = 30$$

Table 5.2

SET		MEASURE			CALCULATE		
Frequency	V _{in} (Vrms)	V _R (Vrms)		(V _C +V _L)(Vrms)	log f	20 log (V _R /V _{in})	20 log (V _C +V _L /V _{in})
0.1 fc1	2Vrms	0.026		2.032			
0.3 fc	2Vrms	0.492		1.902			
0.5 fc	2Vrms	0.933		1.809			
fc1	2Vrms	1.375		1.456			
2fc	2Vrms	1.422		1.399			
3fc	2Vrms	1.478		1.321			
3.5fc	2Vrms	1.583		1.186			
0.5fo	2Vrms	1.613		1.138			
fo	2Vrms	1.955		0.032			
2fo	2Vrms	1.654		1.115			
1700	2Vrms	1.510		1.326			
fc2	2Vrms	1.326		1.474			
2fc2	2Vrms	0.78		1.902			
5fc2	2Vrms	0.279		2.059			
10fc2	2Vrms	0.041		2.055			
12fc2	2Vrms	0.005		2.044			
14fc2	2Vrms	0.025		2.051			
17fc2	2Vrms	0.021		2.009			
20fc2	2Vrms	0.109		1.985			

$$f_{c1} = 28705 \text{ Hz} \Rightarrow \frac{1.955}{\sqrt{2}} = 1.382 \text{ V} \Rightarrow \frac{1}{2\pi \sqrt{LC}} = 234.5 \text{ Hz}$$

$$f_{c2} = 18794 \text{ Hz}$$

$$f_{o(\text{exp})} = 726.6$$

Table 5.3

SET Frequency	MEASURE		CALCULATE	
	V _{in} (Vrms)	V _d (Vrms)	log f	20 log (V _o /V _{in})
	1.04 V _{rms}			
33.65 ← 0.1 fc		2.26		
67.3 ← 0.2 fc		2.19		
134.6 ← 0.4 fc		2.066		
269.2 ← 0.8 fc		1.719		
302.2 ← 0.9 fc		1.632		
336.5 ← f _c		1.594		
437.4 ← 1.2 f _c		1.327		
504.75 ← 1.5 f _c		1.205		
675 ← 2 f _c		0.961		
1346 ← 4 f _c		0.533		
2019 ← 6 f _c		0.329		
3365 ← 10 f _c		0.217		
5047.5 ← 15 f _c		0.144		
6730 ← 20 f _c		0.112		

$$\frac{2.23}{\sqrt{2}} = 1.55$$

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